

STAT/MA 41600
In-Class Problem Set #1: August 26, 2026
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Problem Set 1 Answers

1a. Each outcome in the sample space is a 100-tuple of heads or tails. There are 2^{100} such tuples.

1b. Amongst the 100 flips, if we want exactly 50 flips to be heads (and the others to be tails), there are $\binom{100}{50} = \frac{100!}{50!50!}$ such outcomes.

2a. Consider the 36 possible outcomes; we see that exactly 23 of the 36 outcomes have a sum between 4 and 8 (inclusive); we write them in bold below:

2b. Again consider the 36 possible outcomes; we see that exactly 16 of the 36 outcomes have $|X - Y| \leq 1$; we write them in bold below:

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

3a. Each outcome in the sample space is a 3-tuple of values from r, o, y, g, b, p . There are $6^3 = 216$ such tuples.

3b. The bear family does not contain any red bears if it is a 3-tuple of values from o, y, g, b, p . There are $5^3 = 125$ such tuples.

3c. There are $3! = 6$ “primary” bear families, because each “primary” bear family has r, b, y (in some order). It follows that there are $216 - 6 = 210$ bear families that are *not* primary.

4a. Each outcome in the sample space is a 5-tuple of s, c , or r . There are $3^5 = 243$ such tuples.

4b–4g. For any k with $0 \leq k \leq 5$, to find the number of weekly weather forecasts with k sunny days, there are $\binom{5}{k}$ ways to choose the sunny days, and 2^{5-k} ways to specify the weather on the other $5 - k$ days. So there are $\binom{5}{k} 2^{5-k}$ such weekly weather forecasts. For $0 \leq k \leq 5$, this yields:

4b. $k = 0$: $\binom{5}{0} 2^{5-0} = 32$; **4c.** $k = 1$: $\binom{5}{1} 2^{5-1} = 80$; **4d.** $k = 2$: $\binom{5}{2} 2^{5-2} = 80$; **4e.** $k = 3$: $\binom{5}{3} 2^{5-3} = 40$; **4f.** $k = 4$: $\binom{5}{4} 2^{5-4} = 10$; **4g.** $k = 5$: $\binom{5}{5} 2^{5-5} = 1$.

As in the hint, we also note that $243 = 32 + 80 + 80 + 40 + 10 + 1$.

5a. Using $\int e^{ax} dx = \frac{e^{ax}}{a} + C$, we compute $\int_0^\infty 3e^{-3x} dx = -e^{-3x} \Big|_{x=0}^\infty = 1$.

5b. We compute, using integration by parts:

$$\int_0^\infty (x)(3e^{-3x}) dx = (x)(-e^{-3x}) \Big|_{x=0}^\infty - \int_0^\infty (-1)e^{-3x} dx = 0 - \frac{1}{3}e^{-3x} \Big|_{x=0}^\infty = \frac{1}{3}.$$

5c. Using $e^a = \sum_{x=0}^\infty \frac{a^x}{x!}$, we compute $\sum_{x=0}^\infty \frac{3^x e^{-3}}{x!} = e^{-3} \sum_{x=0}^\infty \frac{3^x}{x!} = e^{-3} e^3 = 1$.

5d. We compute $\sum_{x=2}^\infty \frac{3^x e^{-3}}{x!} = (\sum_{x=0}^\infty \frac{3^x e^{-3}}{x!}) - e^{-3} - 3e^{-3} = 1 - 4e^{-3}$.